

Aryan Thodupunuri

US Citizen | (571) 442-7810 | aryan20544@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Virginia

Charlottesville, VA

B.A in Computer Science · M.S in Data Science | GPA: 3.8/4.0

Expected May 2027

Leadership: Cavalier Autonomous Racing - Perception Team Lead, Association for Computing Machinery - VP of External Outreach

TECHNICAL SKILLS & CERTIFICATIONS

- **Languages:** Python, Java, Go, C++, SQL, JavaScript/TypeScript, Shell
- **Backend & Systems:** FastAPI, Flask, Spring Boot, REST, GraphQL, Microservices, Distributed Systems, Async Processing, PostgreSQL, Redis, DynamoDB, OpenSearch, ChromaDB, Kafka
- **Cloud & Infrastructure:** AWS (Bedrock, EC2, Lambda, S3, RDS, IAM, CloudWatch), Docker, Kubernetes, Terraform, Linux, CI/CD
- **AI/ML:** RAG, Embeddings, Semantic Search, Reranking, TensorFlow, PyTorch
- **Certifications:** AWS Developer — Associate, AWS Solutions Architect — Associate, AWS Machine Learning Engineer — Associate

PROFESSIONAL EXPERIENCE

Amazon Web Services

Arlington, VA

Solutions Architect Intern, Forward Deployed Engineering

May 2026 - August 2026

- Architected a multi-tenant context service for AI coding agents, exposing REST APIs over source code, dependency graphs, runtime state, and team conventions, serving context across **250K+ file repositories at sub-150ms p95 retrieval latency**.
- Built event-driven ingestion microservices across **Lambda, S3, SQS, DynamoDB, and OpenSearch**, processing **1M+ code changes** with incremental indexing and content-addressed caching to cut redundant computation **88%** and indexing costs **63%**.
- Developed an offline evaluation and observability framework across **25K+ coding tasks**, benchmarking retrieval strategies, detecting context regressions with **99.4% precision**, and integrating automated testing into the standard **CI/CD release gate**.

Booz Allen Hamilton

McLean, VA

Software Engineering Intern

May 2025 - August 2025

- Rebuilt a supplier intelligence REST search service in Flask and PostgreSQL, using vector embeddings, query planning and precomputed indexes that reduced p99 latency from **4.6s to 140ms** across **12M+ searchable document fragments**.
- Designed a fault-tolerant contract ingestion engine in Python/C++ with idempotent workers, checkpointing, and dead-letter recovery, processing **500K+ PDFs weekly** with **99.99% completion** while eliminating duplicate writes across distributed jobs.
- Built a distributed indexing service with PostgreSQL data modeling and semantic embeddings, transforming contract data into normalized supplier entities while sustaining **25K+ records per minute** and reducing reprocessing **84%**.

Red Light Management

Charlottesville, VA

Software Engineering Intern

Sept 2024 – May 2025

- Built an event-driven Python ingestion platform processing **5M+ daily records** from Spotify, Apple Music, and Ticketmaster APIs, adding Kafka-style partitioning, idempotent retries, and deduplication to sustain **20K+ events per second** during peak traffic.
- Engineered Redis-backed caching and autoscaled AWS microservices with EC2, Docker, and RDS, cutting **p99 API latency 68%**.

Machine Learning Engineering Intern

April 2024 - August 2024

UVA Link Lab

Charlottesville, VA

- Built a distributed geospatial forecasting pipeline in Python and TensorFlow processing **10M+ sensor observations**, partitioning workloads across parallel workers and caching features to cut training time **94%** and increase experiment throughput **18x**.
- Designed a fault-tolerant model serving API on AWS EC2 with Docker, Redis, REST APIs, and CI/CD sustaining **2K+ prediction requests per second at sub-100ms p95 latency** with automated health checks and zero-downtime rollouts.

TECHNICAL PROJECTS

Relay — Real-Time Group Coordination Platform · Go, PostgreSQL, Redis, WebSockets, React, AWS

- Built a real-time group coordination platform in **Go** serving **300 UVA students**, synchronizing scheduling, rides, expenses, and shared event state over **WebSockets** with **sub-100ms p95** update propagation across concurrent sessions.
- Designed an event-driven collaboration engine with **PostgreSQL transactions, optimistic concurrency, Redis Pub/Sub, and idempotent processing**, resolving competing updates without duplicate bookings while sustaining **10K+ changes per minute**.

CloudSentinel · Python, FastAPI, PostgreSQL, AWS STS, CloudWatch, ChromaDB

- Built a multi-account cloud operations backend in **FastAPI** and **PostgreSQL** that aggregated infrastructure state, cost, and performance telemetry across AWS accounts through **STS AssumeRole** and **CloudWatch**.
- Engineered asynchronous **Python** workers for cross-account discovery and telemetry collection with retries, caching, and failure isolation, combining structured AWS data with **ChromaDB** retrieval to generate contextual cloud recommendations.